

Working with Data in Python Cheat Sheet

Reading and writing files

| Package/Method       | Description                                             | Syntax and Code Example                                                                                                                                                                                                                                                                                               |
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| File opening modes   | Different modes to open files for specific operations.  | Syntax: r (reading) w (writing) a (appending) + (updating: read/write) b (binary, otherwise text)<br><br>Examples: with open("data.txt", "r") as file: content = file.read() print(content) with open("output.txt",                                                                                                   |
| File reading methods | Different methods to read file content in various ways. | Syntax:<br><br>file.readlines() # reads all lines as a list<br>readline() # reads the next line as a string<br>file.read() # reads the entire file content as a string<br><br>Example:<br><br>with open("data.txt", "r") as file:<br>lines = file.readlines()<br>next_line = file.readline()<br>content = file.read() |
| File writing methods | Different write methods to write content to a file.     | Syntax:<br><br>file.write(content) # writes a string to the file<br>file.writelines(lines) # writes a list of strings to the file<br><br>Example:<br><br>lines = ["Hello\n", "World\n"]<br>with open("output.txt", "w") as file:<br>file.writelines(lines)                                                            |
| Iterating over lines | Iterates through each line in the file                  | Syntax:<br><br>for line in file: # Code to process each line                                                                                                                                                                                                                                                          |

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|                    | using a <code>`loop`</code> .                                                               | <p>Example:</p> <pre>with open("data.txt", "r") as file:     for line in file: print(line)</pre>                                                                                        |
| Open() and close() | Opens a file, performs operations, and explicitly closes the file using the close() method. | <p>Syntax:</p> <pre>file = open(filename, mode) # Code that uses the file file.close()</pre> <p>Example:</p> <pre>file = open("data.txt", "r") content = file.read() file.close()</pre> |
| with open()        | Opens a file using a with block, ensuring automatic file closure after usage.               | <p>Syntax:</p> <pre>with open(filename, mode) as file: # Code that uses the file</pre> <p>Example:</p> <pre>with open("data.txt", "r") as file:     content = file.read()</pre>         |

| Pandas         |                                                                    |                                                                                            |
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| Package/Method | Description                                                        | Syntax and Code Example                                                                    |
| .read_csv()    | Reads data from a <code>`CSV`</code> file and creates a DataFrame. | Syntax: dataframe_name = pd.read_csv("filename.csv") Example: df = pd.read_csv("data.csv") |

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| <code>.read_excel()</code> | Reads data from an Excel file and creates a DataFrame.            | <p>Syntax:</p> <pre>dataframe_name = pd.read_excel("filename.xlsx")</pre> <p>Example:</p> <pre>df = pd.read_excel("data.xlsx")</pre>                                                                                 |
| <code>.to_csv()</code>     | Writes DataFrame to a CSV file.                                   | <p>Syntax:</p> <pre>dataframe_name.to_csv("output.csv", index=False)</pre> <p>Example:</p> <pre>df.to_csv("output.csv", index=False)</pre>                                                                           |
| Access Columns             | Accesses a specific column using [] in the DataFrame.             | <p>Syntax:</p> <pre>dataframe_name["column_name"] # Accesses single column<br/>dataframe_name[["column1", "column2"]] # Accesses multiple columns</pre> <p>Example:</p> <pre>df["age"]<br/>df[["name", "age"]]</pre> |
| <code>describe()</code>    | Generates statistics summary of numeric columns in the DataFrame. | <p>Syntax:</p> <pre>dataframe_name.describe()</pre>                                                                                                                                                                  |

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|             |                                                                                                        | <p>Example:</p> <pre>df.describe()</pre>                                                                                                                                                                                                                                                                                        |
| drop()      | Removes specified rows or columns from the DataFrame. axis=1 indicates columns. axis=0 indicates rows. | <p>Syntax:</p> <pre>dataframe_name.drop(["column1", "column2"], axis=1, inplace=True)<br/>dataframe_name.drop(index=[row1, row2], axis=0, inplace=True)</pre> <p>Example:</p> <pre>df.drop(["age", "salary"], axis=1, inplace=True) # Will drop columns<br/>df.drop(index=[5, 10], axis=0, inplace=True) # Will drop rows</pre> |
| dropna()    | Removes rows with missing NaN values from the DataFrame. axis=0 indicates rows.                        | <p>Syntax:</p> <pre>dataframe_name.dropna(axis=0, inplace=True)</pre> <p>Example:</p> <pre>df.dropna(axis=0, inplace=True)</pre>                                                                                                                                                                                                |
| uplicated() | Duplicate or repetitive values or records within a data set.                                           | <p>Syntax:</p> <pre>dataframe_name.duplicated()</pre>                                                                                                                                                                                                                                                                           |

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|             |                                                                                                                                             | <p>Example:</p> <pre>duplicate_rows = df[df.duplicated()]</pre>                                                                                                                                                                                                         |
| Filter Rows | Creates a new DataFrame with rows that meet specified conditions.                                                                           | <p>Syntax:</p> <pre>filtered_df = dataframe_name[(Conditional_statements)]</pre> <p>Example:</p> <pre>filtered_df = df[(df["age"] &gt; 30) &amp; (df["salary"] &lt; 50000)]</pre>                                                                                       |
| groupby()   | Splits a DataFrame into groups based on specified criteria, enabling subsequent aggregation, transformation, or analysis within each group. | <p>Syntax:</p> <pre>grouped = dataframe_name.groupby(by, axis=0, level=None, as_index=True, sort=True, group_keys=True, squeeze=False, observed=False, dropna=True)</pre> <p>Example:</p> <pre>grouped = df.groupby(["category", "region"]).agg({"sales": "sum"})</pre> |
| head()      | Displays the first n rows of the DataFrame.                                                                                                 | <p>Syntax:</p> <pre>dataframe_name.head(n)</pre>                                                                                                                                                                                                                        |

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|               |                                                                                  | <p>Example:</p> <pre>df.head(5)</pre>                                                                                                                                                  |
| Import pandas | Imports the Pandas library with the alias pd.                                    | <p>Syntax:</p> <pre>import pandas as pd</pre> <p>Example:</p> <pre>import pandas as pd</pre>                                                                                           |
| info()        | Provides information about the DataFrame, including data types and memory usage. | <p>Syntax:</p> <pre>dataframe_name.info()</pre> <p>Example:</p> <pre>df.info()</pre>                                                                                                   |
| merge()       | Merges two DataFrames based on multiple common columns.                          | <p>Syntax:</p> <pre>merged_df = pd.merge(df1, df2, on=["column1", "column2"])</pre> <p>Example:</p> <pre>merged_df = pd.merge(sales, products, on=["product_id", "category_id"])</pre> |

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| print DataFrame | Displays the content of the DataFrame.                | <p>Syntax:</p> <pre>print(df) # or just type df</pre> <p>Example:</p> <pre>print(df) df</pre>                                                                                              |
| replace()       | Replaces specific values in a column with new values. | <p>Syntax:</p> <pre>dataframe_name["column_name"].replace(old_value, new_value, inplace=True)</pre> <p>Example:</p> <pre>df["status"].replace("In Progress", "Active", inplace=True)</pre> |
| tail()          | Displays the last n rows of the DataFrame.            | <p>Syntax:</p> <pre>dataframe_name.tail(n)</pre> <p>Example:</p> <pre>df.tail(5)</pre>                                                                                                     |

## Numpy

| Package/Method         | Description                                                                                                                                                                                                                                                                             | Syntax and Code Example                                                                                                                                                                                                                                |
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| Importing NumPy        | Imports the NumPy library.                                                                                                                                                                                                                                                              | Syntax:<br><pre>import numpy as np</pre><br>Example:<br><pre>import numpy as np</pre>                                                                                                                                                                  |
| np.array()             | Creates a one or multi-dimensional array,                                                                                                                                                                                                                                               | Syntax:<br><pre>array_1d = np.array([list1 values]) # 1D Array array_2d = np.array([list1 values], [list2 values]) # 2D Array</pre><br>Example:<br><pre>array_1d = np.array([1, 2, 3]) # 1D Array array_2d = np.array([1, 2], [3, 4]) # 2D Array</pre> |
| Numpy Array Attributes | <ul style="list-style-type: none"> <li>- Calculates the mean of array elements</li> <li>- Calculates the sum of array elements</li> <li>- Finds the minimum value in the array</li> <li>- Finds the maximum value in the array</li> <li>- Computes dot product of two arrays</li> </ul> | Example:<br><pre>np.mean(array) np.sum(array) np.min(array) np.max(array) np.dot(array_1, array_2)</pre>                                                                                                                                               |



# Skills Network



